

ATPL EASA STUDY GUIDE | 070 - OPERATIONAL PROCEDURES

Chapter 7: In-Flight Emergencies: Descent, Evacuation & Ditching

Syllabus Source: CAE Oxford (pp. 235-272) | Pass Target: >= 90% AviationExam

1. EMERGENCY DESCENT (RAPID DEPRESSURIZATION)

A sudden cabin decompression at high altitude (e.g. FL 390) demands immediate flight crew action before Time of Useful Consciousness (TUC < 30 seconds) expires:

- Immediate Memory Items: 1. DON OXYGEN MASKS 100% / Emergency; 2. Establish crew communications; 3. Disengage autothrottle and retard thrust levers to IDLE; 4. EXTEND SPEED BRAKES fully; 5. Pitch down into emergency descent at Vmo / Mmo; 6. Set transponder 7700.
- Target Level: Descend rapidly to 10,000 ft MSL (or FL 100), where oxygen is breathable without supplemental supply, or to the Minimum Sector Altitude (MSA) if flying over mountainous terrain!

2. EMERGENCY EVACUATION STANDARDS

Under CS-25 certification, an airliner must demonstrate complete evacuation of maximum passenger capacity within 90 SECONDS with 50% OF EXITS BLOCKED, using emergency floor lighting and inflatable escape slides.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Emergency Descent Memory Item**

Rapid Depressurization Immediate Actions: 1. Oxygen masks: ON 100% in under 5 seconds, establish crew communications. 2. Pilot Flying (PF): - Thrust levers: IDLE. - Speedbrakes: FULL. - Pitch down to Vmo / Mmo. - Turn 90 degrees off airway to avoid colliding with traffic below. 3. Pilot Monitoring (PM): Squawk 7700, turn on lights, declare MAYDAY to ATC. 4. Level off at 10,000 ft or Minimum Sector Altitude (MSA), whichever is higher.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the event of a rapid decompression at FL 370, what is the VERY FIRST memory action the pilot flying must take?

- [A]: Declare a Mayday to ATC
- [B]: DON OXYGEN MASK (100% / Emergency) and establish crew communications
- [C]: Extend speed brakes
- [D]: Retard thrust levers

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At FL 370, Time of Useful Consciousness is less than 30 seconds. Immediate donning of positive-pressure oxygen masks is paramount to prevent pilot incapacitation.

Q2: Under CS-25 airworthiness regulations, within what maximum time must an airliner demonstrate full passenger evacuation?

- [A]: 60 seconds
- [B]: 90 SECONDS with 50% of available emergency exits blocked
- [C]: 120 seconds
- [D]: 3 minutes

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Evacuation demonstration requires all passengers and crew to evacuate in dark conditions within 90 seconds using only half the exits.

Q3: When ditching an aircraft into open ocean waters with significant swell, in what direction relative to the sea swell should the aircraft be ditched?

- [A]: Directly into the face of the highest swell
- [B]: PARALLEL TO THE SWELL CRESTS (along the swell), touching down along the top of a swell or along the back of a swell
- [C]: Downwind
- [D]: Perpendicular to waves

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Landing parallel to the swell prevents the aircraft burying its nose into the oncoming wall of water, minimizing deceleration forces.